Stellar: Multi-Agent Online Path Discovery for Post-Disaster Stochastic Graphs

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1 Introduction

Natural disasters have a crucial impact on human life [1]. One of the greatest difficulties in disaster recovery processes is the re-establishment of connectivity between regions that were connected before the disaster. Most of the existing literature on stochastic path-finding focuses on average-case shortest-path performance [2],[3], whereas post-disaster routing also requires robustness to severe edge degradation and uncertainty. In this article, we propose *Stellar*, an exploratory multi-agent framework for stochastic path finding scenarios which occur in graphs after a disaster such as earthquakes and network failures.

The main contributions of this work are as follows.

- We propose *Stellar*, a heuristic multi-agent routing framework that combines sampled disaster-aware maps with real-time inter-agent information sharing.
- We show experimentally that heuristic diversity improves the best-arriving agent but can worsen tail performance, while information sharing reduces worst-agent cost by up to 13.5% compared to Baseline.

2 Problem Model

We model the network to be searched as an undirected graph $G = (\mathcal{V}, \mathcal{E})$ where $\mathcal{V}$ is the set of vertices and $\mathcal{E}$ is the set of edges. Each edge $e \in \mathcal{E}$ is associated with a weight constant $w_e \in \mathbb{R}^+$ representing the difficulty of traversing that edge. For the disaster model we use the binary state model described in [4]. After the disaster occurs, an edge can be in two states: affected and unaffected. We denote the state of an edge e by s_e sampled from a Bernoulli distribution $S_e \sim B(\theta_e)$ where $s_e = 0$ is used for unaffected edges, $s_e = 1$ for affected edges and θ_e represents the probability of edge e being affected. The weight of an affected edges is scaled with a cost multiplier function C_e. The cost of traversing edge e after the disaster, W_e is defined as follows:

$$W_e = \begin{cases} w_e & s_e = 0, \\ w_e C_e & s_e = 1, \end{cases}$$

After the disaster occurs, we assume that the states of the edges are unknown. Following a modified version of the CTP definition of Papadimitriou and Yannakakis [5], we assume that the state of an edge can only be discovered during the traversal of the edge. Given a target vertex t, each agent seeks to reach t from the starting vertex s while discovering edge states online. Since multiple agents operate in parallel, we evaluate both the best-arriving agent and the tail of the fleet, measured by ordinal arrival ranks.

3 Stellar Algorithm

To address this online path-discovery problem, we propose *Stellar*, a multi-agent heuristic routing algorithm based on two ideas.

3.1 Disaster Heuristics

Prior work in path-finding and planning shows that problem-specific heuristics can substantially improve search efficiency [6, 7]. Building on this idea, we use a stochastic disaster model, potentially provided by a digital twin of the network [8], to construct agent-specific heuristic maps.

Each agent is equipped with a heuristic map of the graph that represents a prediction of how the disaster affected the original graph. Let $\hat{\theta}_e^{(i)}$ be the heuristic for the probability of e being affected and $\hat{C}_e^{(i)}$ be the heuristic for the cost multiplier when the edge e becomes affected. The heuristic map for agent i, $H^{(i)}$, is constructed as follows:

- Agent i samples the state prediction $\hat{s}_e^{(i)} \sim B(\hat{\theta}_e^{(i)})$, $\forall e \in \mathcal{E}$.
- $H_e^{(i)}$ is defined as: $H_e^{(i)} = \begin{cases} w_e & \hat{s}_e^{(i)} = 0, \\ w_e \cdot \hat{C}_e^{(i)} & \hat{s}_e^{(i)} = 1. \end{cases}$

Using different heuristics based on sampling from a distribution allows agents to explore different paths. This approach is chosen in order to distribute the deviations caused by randomness across agents. If all agents used the same heuristic map, they would tend to select the same route, eliminating the benefit of parallel exploration.

3.2 Information Sharing Between Agents

Unlike single-agent incremental heuristic search methods such as D* Lite [9], *Stellar* uses multiple agents as a parallel search ensemble. During initialization, Stellar launches $n \in \mathbb{N}$ agents where each agent attempts to reach vertex t simultaneously. Each agent creates a route to t by applying Dijkstra's algorithm, assuming its heuristic map $H^{(i)}$ is the real instantiation of the disaster. When an agent starts traversing an edge, its true state is revealed only through traversal. If the edge is affected, the agent incurs the realized cost $w_e C_e$. Otherwise, it incurs w_e. Once the state is revealed, the agent broadcasts this information to the rest of the fleet, allowing other

agents to replan using the updated edge cost. Once a notification about the status of an edge comes, the agents update their heuristic map with the true state, and plan another route. In this way, an edge's true state is discovered once but exploited by all, turning each agent's exploration cost into shared knowledge that improves the whole ensemble's routes.

3.3 Experiment Setup

Experiments use synthetic post-disaster road networks modeled as Manhattan grid graphs. Nine topologies are generated by varying rows and columns over $\{4, 5, 6\}$ (16–36 nodes, 24–60 edges), with the source at the top-left and the target at the bottom-right corner.

Edge costs and damage. Each edge e is assigned a pre-disaster base cost $w_e \sim \text{Uniform}(10, 50)$ and a damage probability $\theta_e \sim \text{Uniform}(0, 1)$, drawn independently. The damage multiplier is drawn per edge as $C_e = 1 + \text{LogNormal}(\mu, \sigma)$, modeling the heterogeneous severity of real infrastructure damage, which is mostly moderate, occasionally catastrophic. We use this distribution not as a calibrated disaster model, but as a controlled way to generate positive, heavy-tailed edge degradation while keeping the median severity fixed across experiments. We fix $\mu = 1.1$, which holds the median damage multiplier at roughly 4×, and sweep $\sigma \in \{0.5, 1.0, 1.5\}$ to vary damage heterogeneity: larger σ lengthens the tail of severe edges while leaving the median fixed.

Conditions. Two fleet sizes are evaluated, $n \in \{5, 10\}$. Under the heuristic conditions each agent independently draws its own severity estimate $\hat{C}_e^{(i)}$ from the same law, knowing the damage *distribution* but not realized values. The three conditions are: Blind, a no-adaptation baseline that follows the pre-disaster Dijkstra path without using discovered edge states for replanning; Baseline, using sampled heuristics with no sharing; and Stellar, the same heuristics with real-time sharing of discovered edge states.

Design. The factorial design gives $9 \times 2 \times 3 \times 30 = 1620$ experiments, each with a distinct seed and all conditions on the same realized graph. Performance is measured at four ordinal ranks by arrival time: agents $\{1, 3, 4, 5\}$ for $n = 5$ and $\{1, 5, 8, 10\}$ for $n = 10$. Significance uses one-sided Wilcoxon signed-rank tests ($\alpha = 0.05$) on paired per-experiment costs.

3.4 Results

Table 1 reports mean route cost by ordinal rank, aggregated over topologies and σ. Markers compare Stellar against Baseline.

Heuristics help the leader but penalize the tail. Sampled heuristics (Baseline) reduce the best agent's cost relative to Blind by 40–48% ($p < 0.001$). However, this benefit does not extend to the rest of the fleet. At the 80th-percentile and worst ranks, Baseline has higher mean cost than Blind, showing that uncoordinated heuristic diversity can benefit the leader while penalizing lagging agents.

Sharing compresses the tail, and the effect scales. Stellar leaves the leader unchanged but significantly improves the 80th-percentile and worst agents over Baseline for both fleet sizes ($p < 0.001$): a state discovered once by any agent becomes free information for all. The effect grows with damage heterogeneity (worst-agent gain rising from 5.4% at $\sigma = 0.5$ to 13.5% at $\sigma = 1.5$, $n = 5$) and with fleet size (the median agent, tied with Blind at $n = 5$, beats it at $n = 10$ for every σ, up to 32.8% at $\sigma = 1.5$).

Table 1: Mean route cost by ordinal rank.

n	Rank	Blind	Baseline	Stellar
5	best	683.9	410.7	413.6
	median	683.9	668.2	654.5***
	80th	683.9	871.4	837.0***
	worst	683.9	1194.0	1064.1***
10	best	726.4	375.4	372.1
	median	726.4	610.0	571.9***
	80th	726.4	906.4	883.2***
	worst	726.4	1336.1	1242.1***

*** $p < 0.001$. Blind cost is constant across ranks.

Scope. Stellar reliably reduces the tail penalty of heuristic routing, but in this regime neither method beats Blind at the 80th or worst ranks, since detours around mostly moderate damage cost more than the damage avoided. Its clear wins over Blind are at the best rank and at the median for $n = 10$; the contribution of sharing is thus variance reduction across the fleet rather than uniform improvement.

3.5 Future Work

Our evaluation is limited to Manhattan grid graphs, which capture regular urban street layouts but not the irregular structure of real road and infrastructure networks. A natural next step is to test Stellar on more complex and realistic topologies to assess how its tail-compression benefit holds under less regular connectivity. The current model also assumes independent edge states, whereas real disaster damage is often spatially correlated, which is left as future work. Further directions include evaluating larger fleets and graph sizes, and studying the effect of delayed or unreliable communication between agents.

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